Parity and invariants

High school · reasoning · Worked solutions · 01

Exercise 1. Start at 7 and add or subtract 4 each move. Can you reach 20? What about 23?

Solution. Every result is congruent to 7 modulo 4, hence to 3. Since 20 is congruent to 0, it is unreachable. Four additions of 4 give 23.

Exercise 2. Five numbers, each 1 or -1, have product -1. Prove that changing two signs per move cannot make all five positive.

Solution. Changing two signs multiplies the product by $(-1)^2 = 1$. The product stays -1, whereas five positive signs give 1.

Exercise 3. Replace two integers a and b by $a + b$. What invariant does this preserve?

Solution. The sum of all integers stays constant: the two terms contributing $a + b$ are replaced by that same number.

Exercise 4. Why need two states of the same parity not be connected by allowed moves? Give an example.

Solution. Parity gives only a necessary condition. With steps of 6 from 0, the target 2 is even but is not a multiple of 6.

Problem · Investigate, justify, conclude. An 8×8 chessboard is coloured alternately black and white. Remove two diagonally opposite corners. Can the 62 remaining squares be tiled by 31 dominoes of size 1×2, each covering two edge-adjacent squares? Count the removed colours and those covered by a domino. Compare with removing two neighbouring corners.

Solution. Opposite corners have the same colour, leaving 30 squares of that colour and 32 of the other. Each domino covers one of each; 31 dominoes would need 31 of each, so tiling is impossible. Neighbouring corners have different colours. A tiling then exists: after removing the two top corners, place three horizontal dominoes in the first row and four in each of the other seven rows. Rotate this construction for the other sides.